THE COHOMOLOGICAL DIMENSION OF A DIRECTED LIMIT OF QUASI-COMPACT QUASI-SEPARATED SCHEMES IS BOUNDED BY THE DIMENSIONS OF THE APPROXIMANTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reconstruction of the sheaf from its pushforwards and the colimit computation of the cohomology on the limit.

Editorial note. The label lemma-cd-limit names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Reconstruction of a torsion sheaf from its pushforwards to the approximants. Same chapter, the lemma stating that a torsion sheaf on the limit is the limit of the pullbacks of its pushforwards along the projections. Supplies the expression F = lim f_i^{-1}f_{i,*}F.
- Cohomology of a directed limit as a colimit. Same chapter, the colimit theorem computing the cohomology of the limit as the colimit of the cohomologies of the approximants. Supplies the identity H^q(X, F) = colim H^q(X_i, f_{i,*}F).

Source reference. Original reference 0EYN: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7722–7722. Prerequisite pointer invoked at proof line(s) 19643; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09YQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7400–7400. Prerequisite pointer invoked at proof line(s) 19646; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $X = \lim X_i$ be a directed limit of a system of quasi-compact and quasi-separated schemes with affine transition morphisms. Then $cd(X) \leq \max cd(X_i)$.*

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Proof. Denote $f_i : X \to X_i$ the projections. Let $\mathcal{F}$ be an abelian torsion sheaf on $X_{étale}$. Then we have $\mathcal{F} = \lim f_i^{-1} f_{i,*}\mathcal{F}$ by Lemma 0EYN. Thus $H^q_{étale}(X, \mathcal{F}) = \operatorname{colim} H^q_{étale}(X_i, f_{i,*}\mathcal{F})$ by Theorem 09YQ. The lemma follows. □